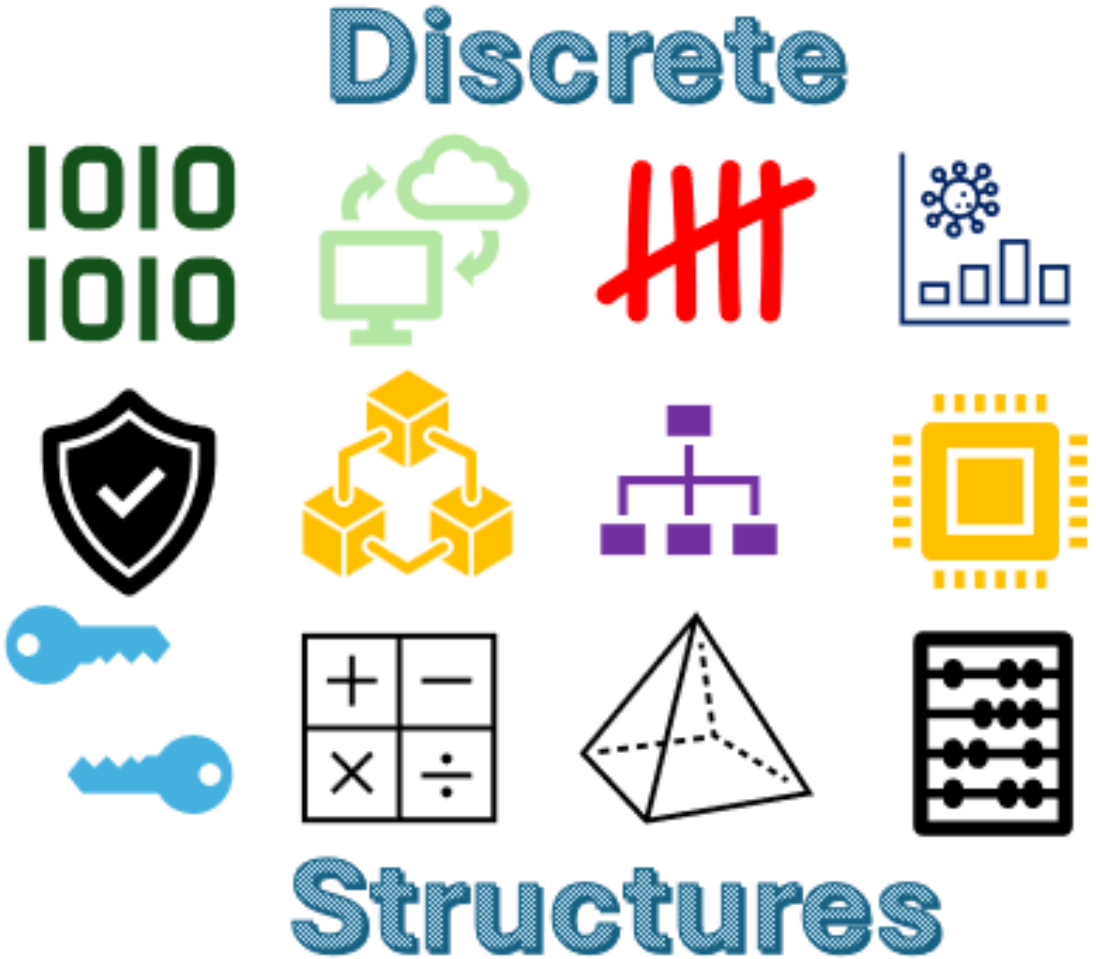
COMP 2310: Discrete Structures
Fall 2026

Lecture 2

Chapter 1: Fundamentals



Today’s agenda

- Chapter 1: Fundamentals
- Practice problems

Operations on sets: Union

- The union of sets A and B is the set of all elements that belong to A or B
- Consider it as concatenation (+)
- $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
- $A = \{a, b, c, e, f\}$, $B = \{b, d, r, s\}$
- **$A \cup B = ?$**
- $A \cup B = \{a, b, c, d, e, f, r, s\}$

Operations on sets: Intersection of sets

- The intersection of sets A and B is the set of all elements that belong to both A and B.
- $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- $A = \{a, b, c, e, f\}$, $B = \{b, e, f, r, s\}$, $C = \{a, t, u, v\}$
- $A \cap B$, $A \cap C$, $B \cap C$
- **$A \cap B = ?$**
- **$A \cap C = ?$**
- **$B \cap C = ?$**
- $A \cap B = \{b, e, f\}$
- $A \cap C = \{a\}$
- $B \cap C = \{\}$

Operations on sets: Complement of sets

- Complement (A - B)
- The complement of B with respect to A is the set of all elements that belong to A but not to B.
- $A - B = \{x \mid x \in A \text{ and } x \notin B\}$
- $A = \{a, b, c\}$, $B = \{b, c, d, e\}$
- Complements of B WRT A, $A - B = \{a\}$
- Complements of A WRT B, $B - A = \{d, e\}$

If U is a universal set containing A, then $U - A$ is called the **complement** of A and is denoted by $\overline{A}$. Thus $\overline{A} = \{x \mid x \notin A\}$.

Operations on sets: Symmetric difference

- What is symmetry of two objects?
- Symmetric Difference ($\oplus$)
- The symmetric difference of sets A and B is the set of all elements that belong to A or B but not both
- $A \oplus B = \{x \mid (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)\}$
- $A = \{a, b, c, d\}, B = \{a, c, e, f, g\}$
- $A \oplus B = \{b, d, e, f, g\}$

Operations on sets: Algebraic properties on set operations

Commutative Properties

1. $A \cup B = B \cup A$
2. $A \cap B = B \cap A$

Associative Properties

3. $A \cup (B \cup C) = (A \cup B) \cup C$
4. $A \cap (B \cap C) = (A \cap B) \cap C$

Distributive Properties

5. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C$
6. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C$

Idempotent Properties

7. $A \cup A = A$
8. $A \cap A = A$

Properties of the Complement

9. $\overline{(\overline{A})} = A$
10. $A \cup \overline{A} = U$
11. $A \cap \overline{A} = \varnothing$
12. $\overline{\varnothing} = U$
13. $\overline{U} = \varnothing$
14. $\overline{A \cup B} = \overline{A} \cap \overline{B}$
15. $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Properties 14 and 15 are known as De Morgan's laws.

Properties of a Universal Set

16. $A \cup U = U$
17. $A \cap U = A$

Properties of the Empty Set

18. $A \cup \varnothing = A$ or $A \cup \{\ \} = A$
19. $A \cap \varnothing = \varnothing$ or $A \cap \{\ \} = \{\ \}$

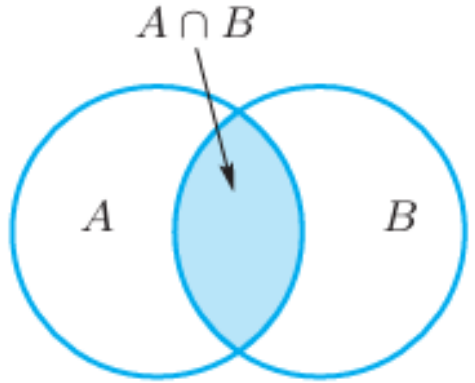
Examples

- See examples 1- 6 in the book

The addition principle

The addition principle

- If A and B overlap, as shown in Figure then elements in A ∩ B belong to both sets
- The sum |A|+|B| counts these elements twice. To correct for this double counting, we subtract |A ∩ B|.
- Thus we have the following theorem, sometimes called the addition principle.
- Because of Figure, this is also called the inclusion-exclusion principle.
- If A and B are finite sets, then $|A \cup B| = |A| + |B| - |A \cap B|$.



A computer company wants to hire 25 programmers to handle systems programming jobs and 40 programmers for applications programming. Of those hired, ten will be expected to perform jobs of both types. How many programmers must be hired?

Solution

Let A be the set of systems programmers hired and B be the set of applications programmers hired. The company must have $|A| = 25$ and $|B| = 40$, and $|A \cap B| = 10$. The number of programmers that must be hired is $|A \cup B|$, but $|A \cup B| = |A| + |B| - |A \cap B|$. So the company must hire $25 + 40 - 10$ or 55 programmers.

The addition principle

- A, B and C are finite sets.
- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$

